



CS 245 Database Management Systems

Lecture #1 Introduction

Rashmi Dutta Baruah

Department of Computer Science & Engineering



भारतीय प्रौद्योगिकी संस्थान गुवाहाटी
Indian Institute of Technology Guwahati

Guwahati - 781039, INDIA





Topics

- Introduction to Databases
- Data Modelling using ER Model
- Relational Data Model and SQL
- ER-to-Relational Model
- Database Design Theory and Normalization

DBMS as a
Black-box

- File Structures, Hashing, Indexing, and Physical Database Design
- Query Processing and Optimization

Internals of relational DBMS

- **Transaction** Processing, Concurrency Control, and Recovery
- **Non-relational DBMS:** NOSQL Systems and Big Data



Books

- **R. Elmasri and S.B. Navathe**, Fundamentals of Database Systems, Pearson, 7th Edition, 2017.
- **R. Ramakrishnan, J. Geherke**, Database Management Systems, McGraw Hill, 2014.
- **A. Silberschatz, H.F. Korth, S. Sudarshan**, Database Systems Concepts, 7th Edition, McGraw-Hill, 2021
- **H. Garcia-Molina, J. Ullman, J. Widom**, Database System: The Complete Book, 2nd Edition, Pearson, 2013.



Grading Components

- End Sem Exam **40%**
- Mid Sem Exam **30 %**
- QUIZ I & II **20 %**
- In-class assignments (in groups of 2) **10%**
- **Dates**
 - Quiz I - 17th February, Monday
 - Quiz II – 21st April, Monday
 - Mid Sem Exam
 - End Sem Exam



Databases

- You may not notice, but you encounter several activities everyday that involve some interaction with a database.
 - Banking and Financial Services, Airline/Railway reservation, Government and Public Services (tax filing) : **Traditional database applications** – stored or accessed information is textual or numeric.
 - Online search engines (Google), Social media platforms (Instagram, LinkedIn), E-commerce platforms (Amazon) : **NOSQL Systems** - stored or accessed information can be documents, programs, images, videos.

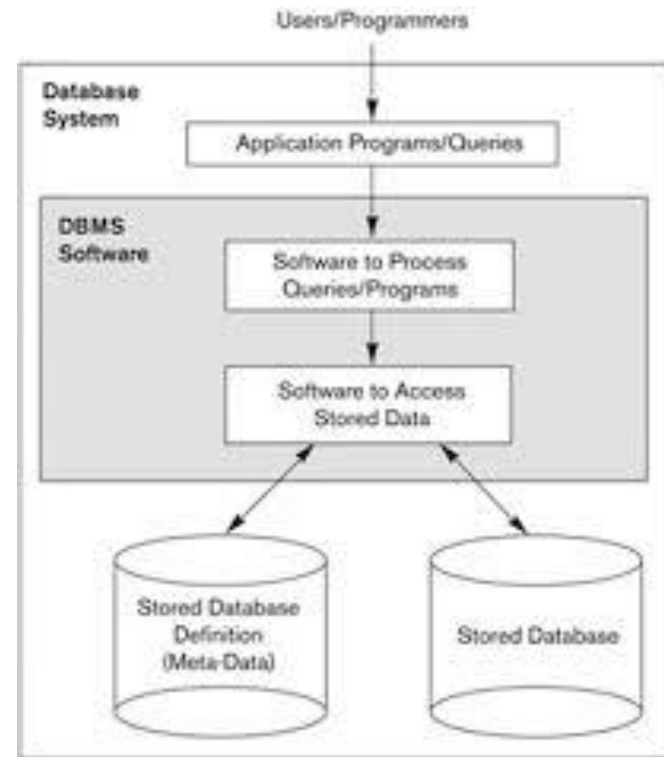


What is a Database?

- Database
 - Collection of related data where data is known facts that can be recorded and have implicit meaning.
- It has following implicit properties:
 - represents some aspects of real-world
 - logically coherent collection of data with some inherent meaning
 - designed, built, and populated with data for a specific purpose

What is a DBMS?

- **Database Management System (DBMS)**
 - general purpose software that enables creation and maintenance of database
- **Database System**
 - The DBMS software together with the data itself. Sometimes, the applications are also included.



A simplified database system environment



Typical DBMS Functionality

- **Define** a database in terms of data types, structures and constraints
- **Construct** or Load the Database on a secondary storage medium
- **Manipulating** the database - querying, generating reports, insertions, deletions and modifications to its content
- **Concurrent Processing and Sharing** by a set of users and programs – yet, keeping all data valid and consistent
- Other features include **protecting and maintaining** the database.



Example of a Database

- **Mini-world for the example:** Part of a UNIVERSITY environment.
- **Some mini-world *entities*:**
 - STUDENTs
 - COURSEs
 - SECTIONs (of COURSEs)
 - (academic) DEPARTMENTs
 - INSTRUCTORs
- **Some mini-world *relationships*:**
 - SECTIONs *are of* specific COURSEs
 - STUDENTs *take* SECTIONs
 - COURSEs *have* prerequisite COURSEs
 - INSTRUCTORs *teach* SECTIONs
 - COURSEs *are offered by* DEPARTMENTs
 - STUDENTs *major in* DEPARTMENTs



Characteristics of Database Approach

- The main characteristics of the database approach versus the file-processing approach:
- **Self-describing nature of a database system**
 - A DBMS **catalog** stores the *description* of the database. The description is called **meta-data**

Relation

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
...	

Columns

Column_name	Data_type	Belongs_to_relation
Name	Char(30)	STUDENT
Student_number	Char(4)	STUDENT
...		



Characteristics of Database Approach

- **Insulation between programs and data**
 - Called **program-data independence**. Allows changing data storage structures and operations without having to change the DBMS access programs.
- **Support of multiple views of the data**
 - Each user may see a different view of the database, which describes *only* the data of interest to that user.
- **Sharing of data and multiuser transaction processing**
 - allowing a set of concurrent users to retrieve and to update the database. Concurrency control within the DBMS guarantees that each **transaction** is correctly executed or completely aborted



Advantages of using Database approach

- Controlling redundancy in data storage and in development and maintenance efforts
- Sharing of data among multiple users
- Restricting unauthorized access to data
- Providing persistent storage for program Objects
- Providing Storage Structures for efficient Query Processing
- Providing backup and recovery services.
- Providing multiple interfaces to different classes of users.
- Representing complex relationships among data.



Advantages of using Database approach

- Enforcing integrity constraints on the database.
- Drawing Inferences and Actions using rules
- Potential for enforcing standards
- Reduced application development time
- Flexibility to change data structures
- Availability of up-to-date information
- Economies of scale



When not to use a DBMS

- **Main inhibitors (costs) of using a DBMS:**
 - High initial investment and possible need for additional hardware.
 - Overhead for providing generality, security, concurrency control, recovery, and integrity functions.
- **When a DBMS may be unnecessary:**
 - If the database and applications are simple, well defined, and not expected to change.
 - If there are stringent real-time requirements that may not be met because of DBMS overhead.
 - If access to data by multiple users is not required.
- **When no DBMS may suffice:**
 - If the database system is not able to handle the complexity of data because of modeling limitations
 - If the database users need special operations not supported by the DBMS.



Summary

- Database - a collection of related data, where data means recorded facts –represents some aspect of real world – used for specific purpose
- DBMS –generalized software package for implementing and maintaining database
- Discussed characteristics of database approach versus file processing applications
- Discussed advantages of DBSM and situations where it is not advantageous to use it